

The empirical recurrence for OEIS A317262, recovered from its tail

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Abstract

OEIS A317262 records “Empirical recurrence of order 67” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 69$. Its last coefficient is $c_{67} = 512 \neq 0$, so no further tail satisfies anything shorter; any order-67 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A317262 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 0, 1, 3, 4, 6, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

5, 17, 21, 74, 158, 440, 1220, 3655, ...

The entry states:

Empirical recurrence of order 67 (see link above).

As of the “Last modified” line on the live entry (Feb 21 2025 14:58:35, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 67: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 67.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 67 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	8	c_{18}	32948	c_{35}	21179545	c_{52}	−2459903
c_2	−25	c_{19}	−18593	c_{36}	−6928040	c_{53}	−11974121
c_3	53	c_{20}	32219	c_{37}	7954504	c_{54}	−8414983
c_4	−110	c_{21}	26725	c_{38}	13245244	c_{55}	−6378089
c_5	170	c_{22}	22075	c_{39}	−18558640	c_{56}	−1265733
c_6	−213	c_{23}	104216	c_{40}	11397713	c_{57}	−396875
c_7	292	c_{24}	−164632	c_{41}	−49740276	c_{58}	1191036
c_8	−280	c_{25}	−92545	c_{42}	10586028	c_{59}	1289494
c_9	−16	c_{26}	241214	c_{43}	−8402522	c_{60}	597424
c_{10}	1695	c_{27}	−1153114	c_{44}	−13432419	c_{61}	−31308
c_{11}	−3504	c_{28}	797309	c_{45}	−25927276	c_{62}	−114424
c_{12}	4425	c_{29}	−1706289	c_{46}	−21613225	c_{63}	−29072
c_{13}	−8832	c_{30}	2840662	c_{47}	47330688	c_{64}	−304
c_{14}	11769	c_{31}	−2167840	c_{48}	35896010	c_{65}	−9152
c_{15}	−14258	c_{32}	−6308252	c_{49}	29504041	c_{66}	1152
c_{16}	2260	c_{33}	7086949	c_{50}	5091893	c_{67}	512
c_{17}	−5185	c_{34}	−5212644	c_{51}	4092326		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 69$, and it is the recurrence OEIS A317262 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{67} - c_1 x^{66} - \dots - c_{67}$. Its constant term is $-c_{67} = -512$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 67 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 67, so $\deg q = 67$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 67, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 67, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 512.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-67 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A317262.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.